



Diversification Of Music Recommendation Systems

by Resham Chana
Student ID: F319472

Loughborough University

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Supervisor: Dr George Vogiatzis
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Abstract

This research project explores how different factors affect how listeners choose music. Different diversity attributes, such as the country of origin and language are explored against popularity of specific songs. Furthermore, Content-Based Filtering Recommendation Systems and Collaborative Filtering Recommendation Systems are tested with the Million Song Dataset in order to diversify listening and recommendations that are given to users. Information Retrieval is carried out to pull meaningful labels from social tagging of songs and K-Means Clustering is implemented to choose songs from different clusters. This is utilised alongside Cosine Similarity techniques to ensure that listeners are interested in the music recommended. Additionally, Singular Value Decomposition is used as a tool for Collaborative Filtering, supplemented with a ranking based on play count for users that are looking for obscurity in listening. The overall aim of this research is to find different methods of diversification to ensure Music Recommendation Systems stay interesting and fresh and allow users to explore the long tail of untapped music.

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3 Introduction

3.1 Project Background

The main focus of this project is Music Recommendation Systems or MRS. The Million Song Dataset will be primarily used as the main data source for exploration and modelling [7]. An MRS is a method for organisations to algorithmically sort through different sources of data to infer customer interests [2]. Web 2.0 has created a paradox of choice where users can be overwhelmed with possibilities, therefore an MRS aims to wade through all the data to present something users are engaged and interested in. The aim is to provide users with exactly what they are looking for when they did not even realise they were searching for it [31].

MRS leverage feedback on items (tracks). This can be ratings on a song or how many times they have played a track. A past interaction from a user and track can be a useful tool when predicting future choices [2].

The Recommender Problem can be formulated in two ways: the prediction version of the problem and the ranking version. The prediction version takes an $m \times n$ matrix, representing m users and n tracks – populated by play count or rankings for each item. The model then predicts the unobserved values of the matrix. The ranking version of the problem chooses the ‘top- k ’ items through different methods [2].

Both of these formulations will be explored in this research project with an emphasis on different goals for the recommendations made. Aggarwal, 2016, proposes four main goals of a recommendation system [2].

These are:

- **Relevance** - This concerns the pertinency and how related the recommendations are to the user.
- **Novelty** – This pertains to the newness of recommendations and the importance of showing an unfamiliar item to a customer. However, in an MRS, the stakes are slightly different as a user is more likely to enjoy repeat listening.
- **Serendipity** - This relates to how surprising the recommendations are to the user. It is the ability of a Recommendation System to potentially aid in discovery of a new genre or artist that they may not have known that they would be interested in.
- **Diversity** - It is important for MRS to provide different types of recommendations. Specifically for a top-k system, if all of the items are very similar, the pitfall then becomes if the user does not care for any of them. Diversity ensures that users are finding dissimilar items as well to prevent boredom.

Another goal for this research is diversity of music in terms of the artists themselves. This includes artists from distinctive countries that speak different languages.

3.2 Main Problems

One of the main issues with MRS is that they may not take into consideration all the aspects that make up a complex taste in music such as: mood, art preference, perceived coolness, culture and diversity. Seaver, 2015, describes grasping the complexities of music taste like “unshuffling playing cards” [31]. The literature regarding capturing cultural aspects is limited – this is a nebulous task and figuring out how music is related to language, nationality and other attributes is a novel area, ripe for exploration.

Another issue is the unexplored “long-tail” of music. Current music consumption is dominated and skewed towards a few mainstream artists and often MRS can homogenise listening towards the head of the long tail [8]. The long tail here represents the less popular and less frequently listened to tracks and artists. Furthermore, when implementing the prediction problem, users are more likely to rate more popular music. This propels the popularity bias and contributes to feedback loops [8].

3.3 Aims and objectives

The main aim of this project will be to create three recommenders.

The first will be a content-based recommender that will leverage social tagging to provide a basis for the diversification of music being recommended. Diversity attributes that describe culture will be defined. These will include: language, country, gender, and users will be able to select whether they would like to expand the music horizons with the addition of these attributes being more varied.

The next recommender will be a collaborative filtering system that ranks music based on play count to ensure users are experiencing more obscure music – which will aid in moving away from the head of the “long tail”.

The third recommender will be used to combine artwork that has similar vibes or colours to recommend music from the same genre. These will encapsulate the overarching goal of diversifying MRS with unique and obscure listening choices and take into consideration culture and art when it comes to computing taste.

Finally, time permitting, the last aim will be to create an application or website to host these MRS.

4 Literature review

4.1 Collaborative Filtering

Collaborative Filtering (CF) is a method to predict preferences for music by understanding how users are likely to have the same tastes if they listen to similar music. This method compares users to each other, and makes recommendations based on the similarity of users. A matrix, M is created of users, u and items, i . The matrix at a point $M_{ua,ij}$ is populated if user, u_a has rated or listened to an item, i_j [8]. The item similarity can then be calculated with Cosine Similarity,

Clustering, Bayesian Belief Nets - just to name a few. The next step in this process is to predict a rating that a user could apply to a song that they have not listened to before [34].

Due to the nature of CF, it is vulnerable to a number of weaknesses. Firstly, there is a cold-start problem wherein it is more difficult to recommend items to users who are yet to listen to or rate music. As well as this, “early-raters” can skew predictions and so previous behaviour can feedback into the system and create a loop. Furthermore, recommendations do not take into consideration the actual content of a song. There is also a popularity bias, as more popular songs are more likely to be rated or listened to and so the propensity to be recommended by the MRS is higher. This creates a “long-tail” of unexplored music [8].

4.2 Content-Based Filtering

Content-Based (CB) Filtering uses information that describes the items to predict user’s preferences. Therefore, subjectivity is excluded, and an objective distance is computed between items to represent the similarity. Common methods for distance include Euclidean, Manhattan and Chebyshev [8].

The information for CB that Spotify uses includes: danceability, energy and valence or music positiveness. Natural Language Processing is also utilised to extract lyrical information and user-created playlist titles can indicate information about the song. If a song appears on a lot of playlists with sad in the title, it is considered a sad song [26].

CB is susceptible to the “grey sheep” problem, occurring when a user has atypical taste and requires diversity of listening. And, when only very similar music is recommended it can bore savant music enthusiasts who are looking for something more novel [8].

4.3 Context-Based Filtering

Context-Based Filtering uses contextual information. Including environmental factors: weather, location, time of day etc. Plus, user-related information such as: emotion, socio-cultural background, activities. As these all can have an impact on the type of music a user would like to listen to [30].

Limitations include the difficulty of measuring and gathering data regarding users, for instance emotions or cultural information. Social tagging can be used but this can be vulnerable to attacks. In 2007, a group of Last.fm users were incorrectly tagging Paris Hilton as “Brutal Death Metal”, leading to her music being recommended to Metal listeners even though she is a Pop artist [20].

Furthermore, if tags are used, the most popular music is more likely to be tagged by users which skews the MRS to the head of the “long tail” [8].

4.4 Hybrid Methods

Hybrid Methods are those that combine different aspects of MRS to mitigate certain limitations [8].

Melville et al. (2002) proposed a Hybrid Method to tackle sparsity and the “early-rater” problem. A CF algorithm is boosted using Naïve Bayes as a content classifier. Predictions for ratings are made based on features such as career, sex, age of user. This hybrid model improves upon a sole Content-Based Model or a Collaborative Model [23].

4.5 Popularity Bias

MRS can often be bias towards the head of the “long-tail”. This is mentioned extensively within the 2010 book “Music Recommendation and Discovery” by Oscar Celma [8]. The source used to showcase the consumption of popular music is the 2007 “State of the Industry” report [22]. This declares that in the United States and Canada in 2008, the top 5 albums accounted for 33% of all album sales. Only the United States and Canada is reported on, which is not representative of the entire music industry and this source is now dated.

In the 2019 article “The Unfairness of Popularity Bias in Recommendation” Abdollahpouri states “a downfall of CF is popularity bias”, referencing recommendation systems in general and not specifically MRS [1].

Kowald et al., 2020 reproduced Abdollahpouri’s work for MRS. A Last.fm dataset of 1 billion listening events was used to analyse a user’s tendency to popular music. Only 20% of Last.fm users listen to unpopular artists. Furthermore, with baseline approaches, KNN-based approaches, and Matrix Factorization-based MRS, more popular items are recommended more often than unpopular

ones. Plus, users with more obscure taste receive worse recommendations than those with popular tastes [19].

To mitigate popularity bias within MRS, Zhang et al. (2021) outlines several methods: boosting less popular items with ranking adjustment; adjusting weights for training data using inverse propensity scoring and casual embedding which randomises items [41].

4.6 Mainstreamness and Diversity of Listening

A seminal paper by Bauer and Schedl investigates the “Mainstreamness” of MRS. Asserting that using popularity of music is a common practice within MRS. Nevertheless, listeners with specialised tastes are left unsatisfied. Therefore, to alleviate this, a rank-based approach is used to decrease bias towards the long tail. This method uses popularity of music within the country of the user. This also takes into account the varied cultural perceptions of a listening experience. However, using country of origin as a proxy for culture can disregard diasporas and multicultural cities. Moreover, using this methodology might in fact emphasise the mainstream music in a specific country rather than increase diversity of listening [5].

Another interpretation of “diversity” is the similarity between songs that are recommended. A 2020 paper from Anderson et al. (2020) found that Spotify encourages homogenisation of listening habits. Users are more likely to have diverse listening without an MRS. Unlike the other research papers discussed, Anderson et al. used actual Spotify data [3].

Diversity can also refer to the types of artists being recommended. Some research has found that Spotify is likely to gender certain genres [11]. For instance, rock is seen to be male dominated and so recommendations are probable to be male. Therefore, this reinforcing of stereotypes within music can be seen as upholding institutions of [40]. Other research has also found gender biases in MRS. Eriksson and Johansson (2017) found a majority of Spotify bot users were recommended male artists. However, the algorithms used by Spotify are constantly changing and so this study may not be currently applicable [12].

4.7 Album Artwork

Finally, the research conducted for this specific project will also be looking at Album Artwork and genre. Unfortunately, there is a gap in research here and the limited information regarding this topic was found on a blog [6]. On this blog, the author describes the visual proxy that album artwork serves for the sound of a song. However, from the sample of albums studied, there was not a strong correlation between the similarity of a song and the similarity of the artwork.

4.8 Gaps

The gaps within the research include the lack of information on how album artwork can influence what people want to listen to. In playlists such as the Spotify playlist “Red Music”, all of the music saved has red album artwork. This playlist has 1,872 “likes” demonstrating that this is a feature consumers desire [37]. Another gap would be the data used. The actual streaming platforms themselves are not well studied from an academic perspective and there is a lack of open dialogue between the consumers and the companies.

Another gap would be within the “diversity” category. Computer Science research often refers to the diversity of music recommendations in terms of computational similarity. But biases around race, class, sexuality, language and religion are not well studied in the context of MRS.

4.9 Seminal Works

Seminal works include the research into Popularity Bias from Kowald et al., 2020 [19]. As well as the research into “Mainstreamness” from Bauer, C. and Schedl, M., 2019 [5]. Moreover, the first CF approach was used with Tapestry which filtered emails based on reactions to documents. This project coined the term Collaborative Filtering [15].

4.10 Conclusion

In conclusion, the exploration of Music Recommendation Systems illustrates the balance between providing personalised music suggestions and reducing certain limitations such as: cold-start, grey-sheep and feedback loops.

Furthermore, issues such as popularity bias, “mainstreamness” of recommendations, and challenges in accurately incorporating contextual factors like culture can affect the quality of MRS. Moreover, gaps in understanding, particularly regarding the influence of album artwork on music preferences and the impact of biases related to cultural context highlight an area that further research would illuminate.

In summary, while MRS represent a powerful tool for navigating the musical landscape, continued research is necessary to ensure that these systems meet the needs of users and contribute to a more inclusive and diverse musical ecosystem.

5 Methodology

5.1 Data Collection

5.1.1 Million Song Dataset - Track Metadata

The first dataset that has been used for this project is a part of the “Million Song Dataset” or MSD project [7]. The metadata from this vast dataset will be the basis for analysis within this project. The track metadata has 1,000,000 songs and has the following features:

- `track_id` – Track ID from Echo Nest
- `title` – Song Title
- `song_id` – Song ID from Echo Nest
- `release` – Album Name
- `artist_id` – Artist ID from Echo Nest
- `artist_mbid` – ID from musicbrainz.org
- `artist_name` – Artist Name
- `duration` – Duration in Seconds
- `artist_familiarity` – Algorithmic estimation of how well known the artist is
- `artist_hottnesss` – Algorithmic estimation of how much “buzz” the song has
- `year` – Song Release Year
- `shs_perf` – Performance number on secondhandsongs.com
- `shs_work` – Clique Numbers from secondhandsongs.com test and train files

There are 1,000,000 unique tracks and there are no null values. The metadata contains 13 of the attributes from the wider dataset from MSD. The wider dataset contains the attributes found in found in Table 1. The entire MSD dataset is 280 GB, which is unfortunately too large for standard computer memory and so the metadata is chosen for this project. There is also a subset of 10,000 songs of the MSD which contains the full set of features and is 1.8 GB.

Although there are fewer attributes in the metadata, there are a greater number of tracks than the subset and as the aim of the project is to diversify MRS it is considered more important to have a greater number of tracks.

5.1.2 Last.fm Tags Dataset

The next dataset is from the same source and utilises data from Last.fm. Last.fm is a website where users can connect to Spotify, YouTube, Apple Music, Tidal and Soundcloud to track music. It is used as a music discovery tool and data collection service. Moreover, Last.fm can be used to manually tag music with descriptions, such as: **Female Vocalist**, **Ethiopian Jazz** or **90s** [20]. The Last.fm Tag Dataset has 8,598,630 rows and the following features [7]:

- `tid` – Track ID which matches to the MSD track ID
- `tag` – User created description of the track
- `tag_number` – Count of tags for each track

The dataset contains 522,366 unique tags, 505,216 unique tracks and no nulls.

Table 1: Table of MSD attributes [7]

Field Name	Type	Description
analysis sample rate	float	sample rate of the audio used
artist 7digitalid	int	ID from 7digital.com or -1
artist familiarity	float	algorithmic estimation
artist hotttnesss	float	algorithmic estimation
artist id	string	Echo Nest ID
artist latitude	float	latitude
artist location	string	location name
artist longitude	float	longitude
artist mbid	string	ID from musicbrainz.org
artist mbtags	array string	tags from musicbrainz.org
artist mbtags count	array int	tag counts for musicbrainz tags
artist name	string	artist name
artist playmeid	int	ID from playme.com, or -1
artist terms	array string	Echo Nest tags
artist terms freq	array float	Echo Nest tags frequencies
artist terms weight	array float	Echo Nest tags weight
audio md5	string	audio hash code
bars confidence	array float	confidence measure
bars start	array float	beginning of bars, usually on a beat
beats confidence	array float	confidence measure
beats start	array float	result of beat tracking
danceability	float	algorithmic estimation
duration	float	in seconds
end of fade in	float	seconds at the beginning of the song
energy	float	energy from listener point of view
key	int	key the song is in
key confidence	float	confidence measure
loudness	float	overall loudness in dB
mode	int	major or minor
mode confidence	float	confidence measure
release	string	album name
release 7digitalid	int	ID from 7digital.com or -1
sections confidence	array float	confidence measure
sections start	array float	largest grouping in a song, e.g. verse
segments confidence	array float	confidence measure
segments loudness max	array float	max dB value
segments loudness max time	array float	time of max dB value, i.e. end of attack
segments loudness max start	array float	dB value at onset
segments pitches	2D array float	chroma feature, one value per note
segments start	array float	musical events, note onsets
segments timbre	2D array float	texture features (MFCC+PCA-like)
similar artists	array string	Echo Nest artist IDs (sim. algo. unpublished)
song hotttnesss	float	algorithmic estimation
song id	string	Echo Nest song ID
start of fade out	float	time in sec
tatums confidence	array float	confidence measure
tatums start	array float	smallest rhythmic element
tempo	float	estimated tempo in BPM
time signature	int	estimate of number of beats per bar, e.g. 4
time signature confidence	float	confidence measure
title	string	song title
track id	string	Echo Nest track ID
track 7digitalid	int	ID from 7digital.com or -1
year	int	song release year from MusicBrainz or 0

5.1.3 Echo Nest Triplets

The next dataset used is also from the MSD source. The dataset contains user listening history with a play count for each track. The dataset has 48,373,586 rows, 384,546 unique MSD songs, 1,019,318 unique users and no nulls. It has the following columns [7]:

- **user** – unique ID for each user
- **song** – Song ID that matches the MSD song ID
- **play_count** – Play Count for each song for each user

This dataset can then be used to create a grouped dataset that computes the total play count of each song.

5.1.4 Genre Dataset

A genre dataset is also provided by MSD where a categorisation of genre is extracted using information from allmusic.com. The columns in this dataset are [24]:

- **track_id** – MSD track ID
- **genre** – Song Genre

This dataset has 422,714 unique tracks that are labelled with their genre. This is displayed in Table 2.

Table 2: Number of Tracks per Genre

Genre Name	Number of Tracks
Pop/Rock	238,786
Electronic	41,075
Rap	20,939
Jazz	17,836
Latin	17,590
R&B	14,335
International	14,242
Country	11,772
Religious	8,814
Reggae	6,946
Blues	6,836
Vocal	6,195
Folk	5,865
New Age	4,010
Comedy/Spoken	2,067
Stage	1,614
Easy Listening	1,545
Avant-Garde	1,014
Classical	556
Children	477
Holiday	200

5.1.5 Image Dataset

Another dataset has been created from the MSD tracks [25] by Oramas et al. which captures the album artwork of 30,713 MSD tracks. These images create the foundation for the more artistic investigation of music and the relation to drawings, photographs and colours. The columns within this dataset are as follows:

- **msd_track_id** – MSD track ID
- **genre** – Song Genre
- **album_index**
- **set** – test, train or val
- **image_url** — The download link for each album artwork

5.2 Data Cleaning and Processing

Firstly the data is processed using Python so that it is in a usable state. The Echo Nest Triplets are grouped into tracks to ascertain the total play count. This feature is key when characterising obscure music for the “savant” users. The Image Dataset only included URLs for album artwork, so these are downloaded with the `requests` package [29]. Images for songs from the same album are skipped to avoid duplication and skewing of prediction models. The code snippet to perform this is as follows:

```
def download_image(url, path):
    '''
    Funtion to download images from a URL
    '''
    try:
        response = requests.get(url, stream=True, timeout=10)
        response.raise_for_status()
        with open(path, 'wb') as file:
            for chunk in response.iter_content(1024):
                file.write(chunk)
    except Exception as e:
        print(f"error: {e}")

# base directory for the images
base_dir = 'C:\Users\resha\images'

# keep track of downloaded URLs
downloaded_urls = set()

# iterate through the dataset and download images
for index, row in images_df.iterrows():
    set_name = row['set']
    genre = row['genre']
    url = row['image_url']
    # skips the duplicates if already downloaded
    if url in downloaded_urls:
        print(f"Skipping already downloaded URL: {url}")
        continue
    # create directory
    dir_path = Path(base_dir) / set_name / genre
    dir_path.mkdir(parents=True, exist_ok=True)
    # Define the image path
    image_path = dir_path / f"{row['msd_track_id']}.jpg"
    download_image(url, image_path)
    downloaded_urls.add(url)
```

The track metadata database is converted to a Pandas DataFrame object using the `SQLite` library. Also, the relevant columns are kept, whilst features that are not used are deleted. The Last.fm Tag Dataset is also converted to a Pandas DataFrame.

6 Data Analysis

6.1 Million Song Dataset - Track Metadata

The Track Metadata provides useful information about the year of release for songs within this dataset. There are a number of values set to 0 within the year column. These are removed, leaving 515,576 rows. This visualisation can be seen in Figure 1. As demonstrated, a greater number of tracks are released in later years, with the most being released between 2000 and 2010. There is a steep drop with fewer tracks being released in 2011 due to the year of publication for this data.

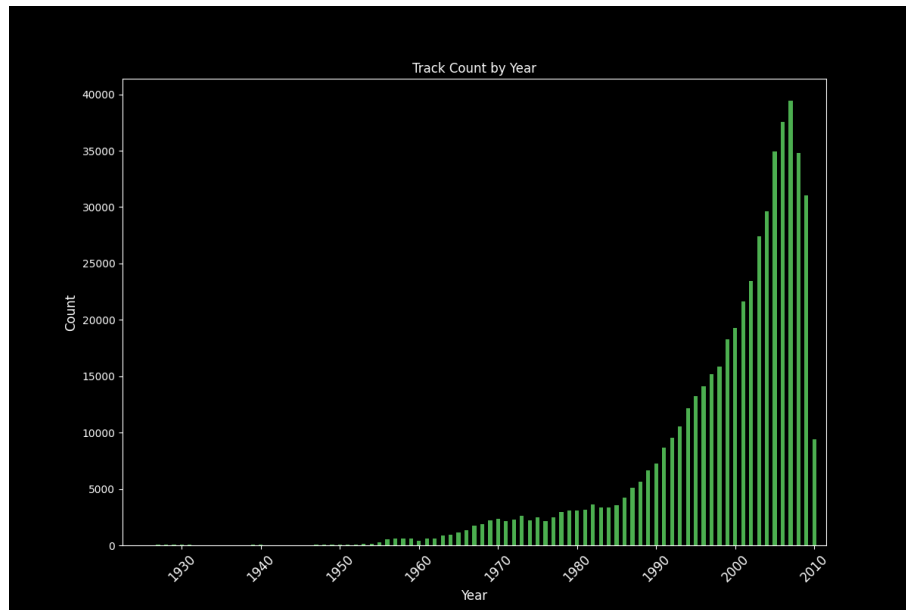


Figure 1: Year Distribution

6.2 Last.fm Tag Dataset

The Last.fm Tag Dataset is explored to visualise and understand the contents of the user submitted tags. The tags provide interesting insight into how people label the music they are listening to. The first visualisation shown in Figure 2 displays the top 20 tags within the data and the count of each.

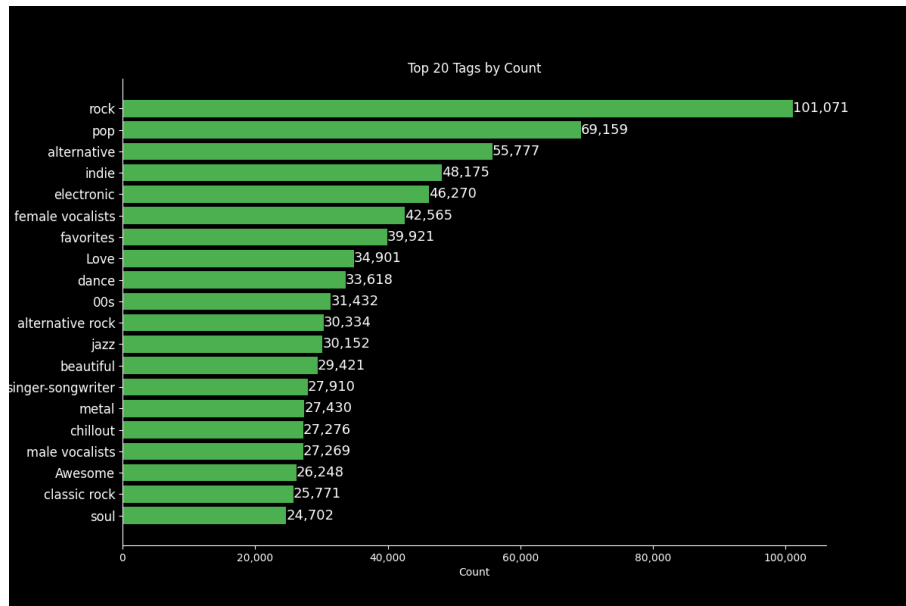


Figure 2: Most Popular Tags

As demonstrated in Figure 2, the tags are varied and encapsulate the geographic location of artists as well as genre, year and gender. Moreover, there are multiple tags that could harbour similar meanings. This includes “female” and “woman”, or “British” and “UK”. The tags are cleaned by converting the column to string types, removing special characters and any tags where their length is less than 1. The number of unique tags reduces to 518,722.

6.2.1 Gender Grouping

To encapsulate the diversity of the music within this dataset, different tags can be utilised. Experiments are carried out to pull out meaningful features within the tags. Features can be extrapolated

with matches: for example a new column “geographic” can be created with matches of strings such as “British”, “American” etc. Although, this would not capture strings such as “Britain” or “Americana”. Consequently, a fuzzy matching can be employed which produces a similarity score. The python package `thefuzz` [28] uses the Levenshtein distance [21]:

$$lev_{a,b}(i,j) = \begin{cases} \max(i,j) & \text{if } \min(i,j) = 0 \\ \min \begin{pmatrix} lev_{a,b}(i-1,j) + 1 \\ lev_{a,b}(i,j-1) + 1 \\ lev_{a,b}(i-1,j-1) + 1 \end{pmatrix} & \text{otherwise} \end{cases}$$

This is a function that computes the number of minimum changes to transform one string into another. These changes include replacements, insertions or deletions for each character in the string. A matrix is used with columns and rows representing the characters in the strings and edit distances are calculated for each substring in an iterative fashion. **thefuzz** package implements a number from 0 to 100 where 100 indicates an exact match. Within the context of the Last.fm tags, each string can be compared to every other string and output a distance. Strings above a certain threshold indicate a match. However, the time complexity of comparing every tag to every other tag is $O(n^2)$ where n is the number of tags. A loop is written to compare each tag and create groupings based on a matching of over 90, however this experiment is deemed inconclusive as it cannot run in its entirety due to the time complexity.

The next experiment is to only compare the group for gender. A new column “gender” is created wherein if the tag is a string where the Levenshtein Distance indicates a close proximity to the strings “male” or “man”, then the gender column is populated with “male”. The functions `fuzz.partial_ratio()` and `fuzz.ratio()` are experimented with, as well as the threshold for matches.

When a `fuzz.partial_ratio()` match is executed for “man” and “german” the score is 100. This results in incorrectly labelled tracks. Therefore, a more strict fuzzy match is implemented with a score tracker and whichever score is highest for male or female matching it is applied to that tag. This improves accuracy slightly, but without a ground truth for gender, it is difficult to ascertain how precise these results are. Figure 3 shows the word cloud for tags that are categorised as female using this method.



Figure 3: Tags Labelled as Female

Moreover, bands may not be tagged with a gender and so could be missed out of this labelling. From 505,021 unique tracks this method results in 19% being labelled as either male or female. It is also important to note that non-binary artists have been missed out of this analysis. This would be an opportunity for further research.

6.2.2 Geographic Grouping

Another tag that could increase the level of diversification would be geographic tags. These are pulled out of the data using a similar method as the male and female labels. There are several categories for geographical groupings. These are split up into continent, nationality, language and religion. The CIA World Factbook [10] is used as a database for countries and the corresponding nationalities. This is processed into a list of tuples and matched to the tags column within the

Last.fm Dataset. Again, a fuzzy matching is implemented and a new column **nationalities** is created. The CIA World Factbook is again used to find most spoken languages and match the tags to these. For religion, the most populous were chosen for their abundance within the tags [9]. Finally, the continents were extrapolated using the same fuzzy matching technique.

The data is then pruned. The mean count per tag is 16.58, i.e. the number of times that tag is used. Therefore, tags that are used less than or equal to 17 times are removed from the dataset. The number of rows reduces from 8,570,669 to 7,274,424 after pruning. This ensures solely valuable tags are selected using the **num-apps** technique, which evaluates tag quality based on application across all items [32]. Furthermore, a pivoted dataset is created so that each tag assigned to a track has a dedicated column. This ensures each track is unique and without duplicates so that it can be joined with other features within other datasets.

6.3 Merged Dataset

6.3.1 Pairwise Plot

A merged dataset is created using: the Last.fm Tags; Track Metadata; the play count retrieved from the Echo Nest Triplets and the Genre Dataset. Moreover, the NaN rows are dropped, which results in 151,029 rows in the DataFrame. A pairwise plot, using the **seaborn** [39] package, is created. To account for the skewness of the play count, a log transformation is applied to this feature.

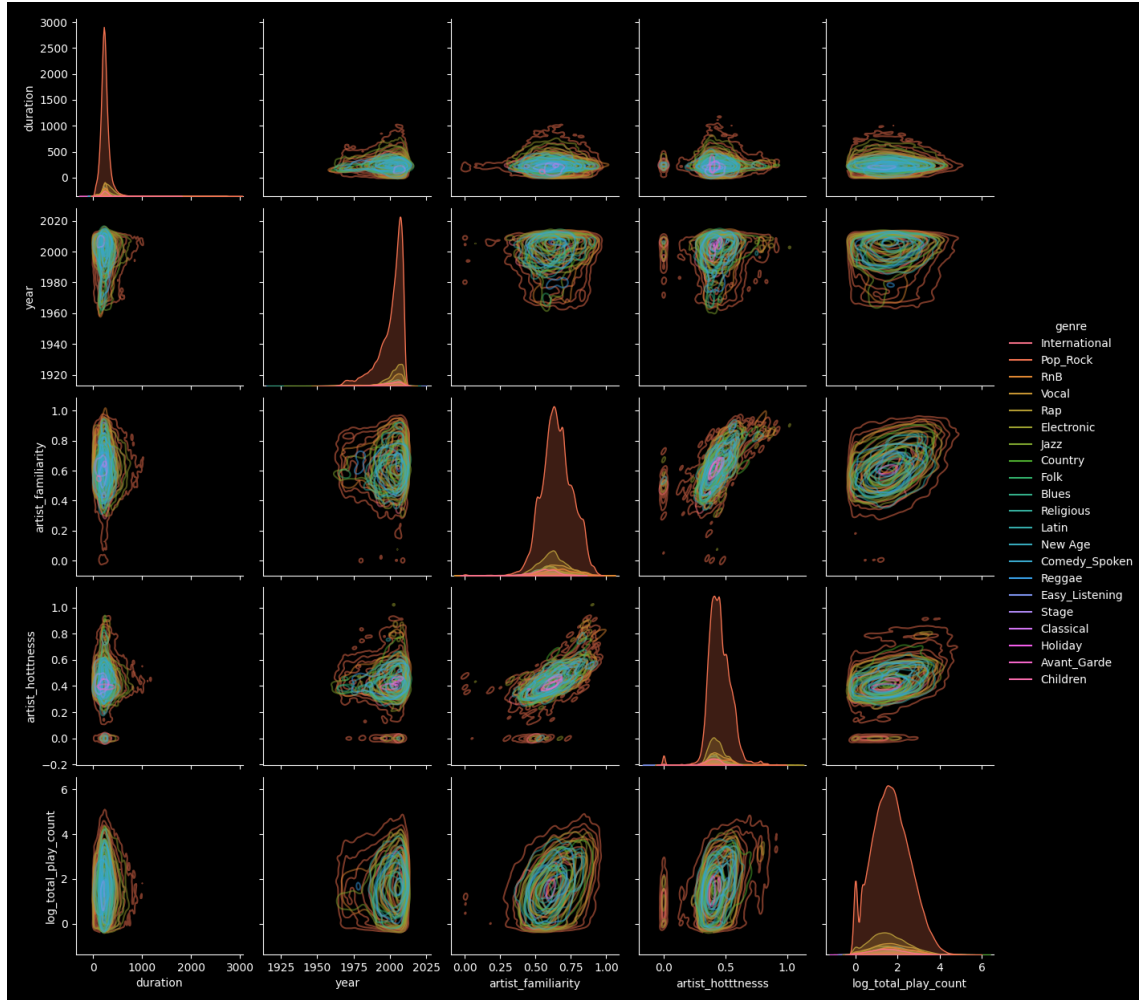


Figure 4: Pairwise Plot of different features

The plot in Figure 4 demonstrates that there is a small variability in the duration of the tracks. The year distribution is skewed towards the 2010s, with a large proportion being Pop.rock music. There is a weak positive correlation between artist familiarity and play count. Another point to note is that there seems to be a significant number of Country music played before the 1980s, as well as R&B music in the 1990s. As well as this, it is clear that how “hot” or “familiar” an artist

is deemed, is positively correlated with the popularity of the track. This is an interesting aspect to note as leaning into these features for recommendations could hinder the discovery of music within the “long tail”. Using these features is a recipe for leaving behind obscure and peculiar music for savant listeners.

6.3.2 Gender and Population

Next, the gender column is joined from the fuzzy matching which can be inspected to find the distribution of play count. This is displayed in Figure 5. This shows how the female tagged tracks have a median higher play count than the male tagged tracks. Furthermore, the female tagged tracks have a greater dispersion of data with an outlier showing a large play count.

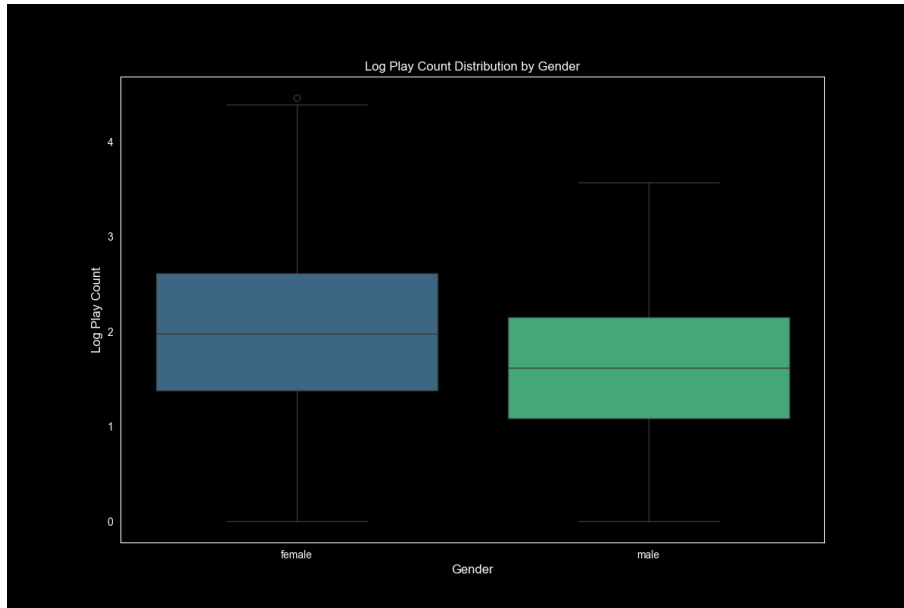


Figure 5: Gender Play Count Box plot

6.3.3 Gender by Genre

As mentioned in section 4.6, Spotify recommendations can gender certain genres like Rock being dominated by males. This notion is explored in Figure 6.

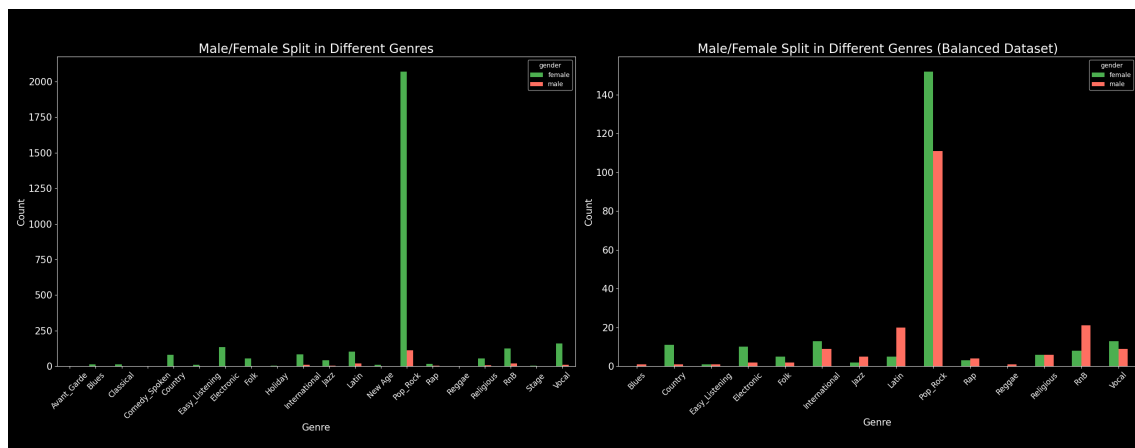


Figure 6: Number of Tracks by Gender and Genre

Females account for 92.71% of the tracks and so there is an overwhelming number of females that are captured in this first plot. The gender is pulled from the social tags and so it is fascinating that most tags associated with gender are female.

As more users engage in tagging with features they care about, this could suggest that users are interested in the diversification of music by gender.

The data is re-balanced and a smaller subset is plotted, which can be seen next to the original plot. In the balanced set there are more males in genres like R&B, Jazz, Rap and Latin. This demonstrates the “gender-isation” of these genres in the dataset as discussed within the Literature in Section 4.6. A diverse recommender could shift away from the norms presented into more balanced suggestions.

6.3.4 Country and Play Count

After this, the country column is joined from the fuzzy matching process. This results in 7272 unique tracks with a country label and play count. This is a greatly reduced number of tracks from the original dataset and it would be optimal if this information was already available. The MSD subset - containing 10,000 songs - is explored as it contains the location attribute shown in Table 1.

The location field is also quite disorganised with US states, Countries and Cities. This is matched using the CIA Factbook data [10] and a new column `country` is created. There are 2283 tracks with valid locations and play count.

This dataset is more reliable than using locations matched from tags. The trustworthiness of tags could be questioned as they are user submitted [8]. For example, Bryan Adams has been incorrectly tagged as an artist from Eswatini with the fuzzy matching process, when he is actually a Canadian artist [16].

Therefore, the subset MSD data is used to demonstrate the geographic distribution. The dataset with the 2283 tracks is used - which encapsulates the MSD attributes, location and play count. The sum of the log play count for each track, grouped by country, is plotted. This is shown in Figure 7. The plot was created using `plotly` [17] and `coco` [33].

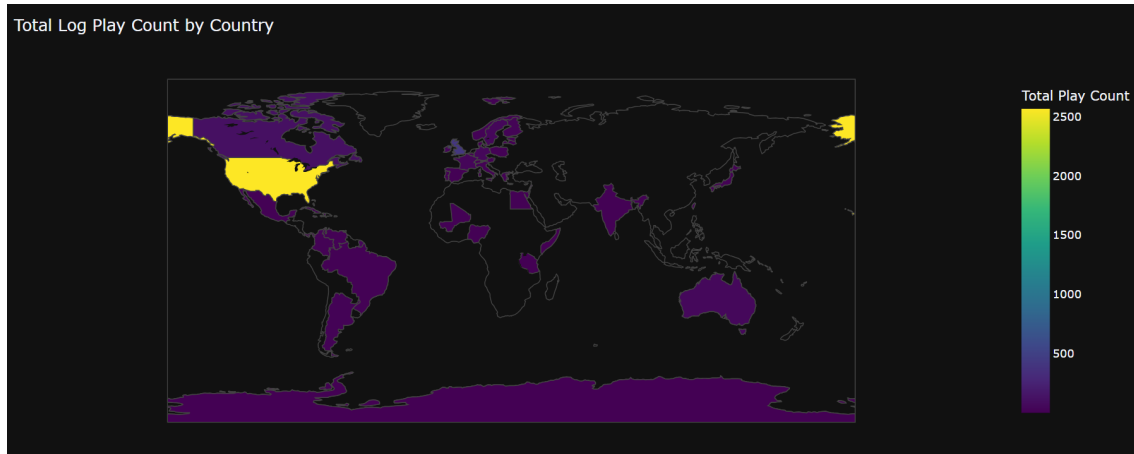


Figure 7: Map displaying Play Count by Country

Figure 7 demonstrates the large skew of play count to tracks from the United States. The popularity bias of many MRS stated in section 4.5, suggests that a standard MRS would endorse more music from the United States and United Kingdom which would decrease the diversity of listening.

6.3.5 Overall Counts from Fuzzy Information Retrieval

For the remainder of the research the MSD subset is not utilised. Even though the accuracy of the countries may be increased, this is not a certainty. Furthermore, the subset still does not provide ground truth for gender or play count. Therefore, the merged dataset is used and explored.

In Table 3, there are the top 10 countries by count in the merged dataset. United Kingdom is the most popular tag label that has been matched.

In Table 4, there is the gender distribution with the overwhelming number of female tagged tracks. Again, this poses the question of why users are more likely to tag a song with female than male.

Table 5 shows the top 10 languages tagged with English interestingly not at the top, even though United Kingdom and United States are the top countries. This could be because many tracks have the implication of being in English and tags are used for more novel aspects of the song.

Country	Count
United Kingdom	1942
United States	1382
World	1219
Spain	1197
France	968
Germany	687
Italy	615
Sweden	587
Mexico	440
Finland	416

Table 3: Top 10 Countries by Count

Gender	Count
Female	5163
Male	406

Table 4: Gender Distribution

Language	Count
Spanish	1161
French	898
Italian	763
German	659
English	353
Japanese	168
Chinese	155
Portuguese	111
Russian	95
Turkish	85

Table 5: Top 10 Languages by Count

Religion	Count
Christianity	1512
Paganism	102
Judaism	36
Rastafari	26
Islam	12
Hinduism	7
Buddhism	1

Table 6: Religion Distribution

Table 6 displays the number of religious tags, with Christianity as the most popular religion tag.

Continent	Count
North America	1170
South America	967
Africa	412
Europe	99
Australia	99
Asia	64
African	5

Table 7: Continent Distribution

Finally, Table 7 demonstrates the distribution for different Continents. This result is expected from the top languages being English and Spanish, with many songs being tagged as “Latin”.

6.4 Image Dataset

The image dataset from described in Section 5.1.5 is explored next. From the discussion in Section 4.7 it is clear that there is a gap in the literature in apropos to how an album artwork can convey a ‘vibe’ or feeling for a specific song or genre.

The first aspect that is investigated is the dominant colour of each artwork. The package `ColorThief` is used to extract the dominant colour from each song’s album artwork [13].

```
def extract_main_colour(filepath):
    """Function to extract the main colour from an image"""
    color_thief = ColorThief(filepath)
    dominant_colour = color_thief.get_color(quality=1)
    return dominant_colour
```

This outputs three values for the RGB or Red, Green, Blue segments of the colour. The dominant colours for each genre are plotted in a 3-D visualisation. The Jazz and Metal songs are demonstrated in Plot 8. These were both chosen to compare as they have a similar number of tracks assigned to these labels in the dataset. Clearly, Jazz tracks have art with a wider range of colours than Metal. Also, the Metal artwork appears to be darker than Jazz, as it has less concentration around the white and light colours. This suggests that artwork could in fact give us some information about the genre of a track.

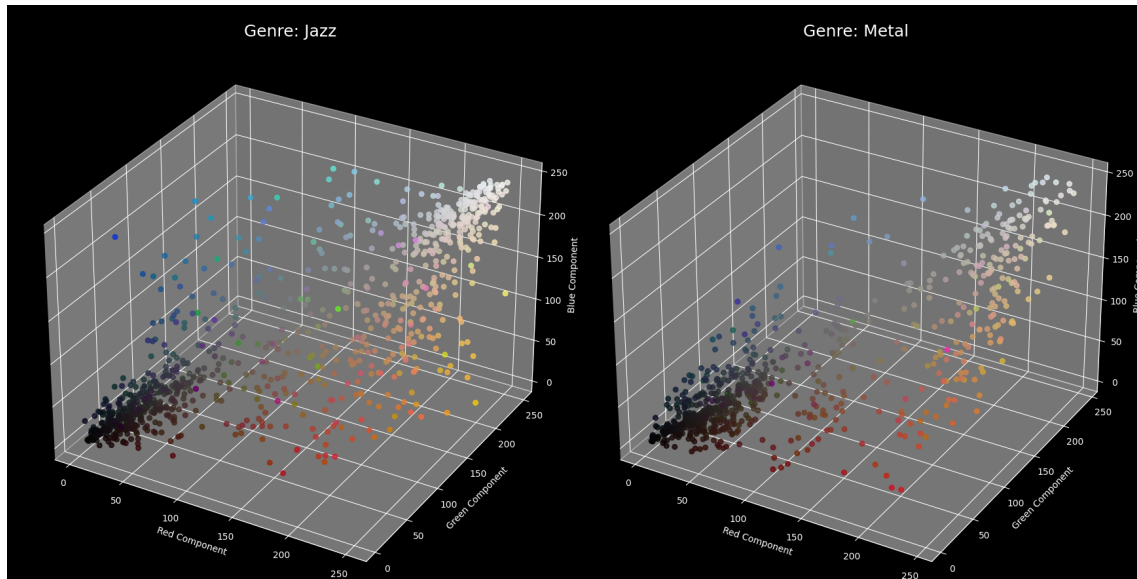


Figure 8: The Dominant Colours within Jazz Music and Metal Music Album Artwork

7 Modelling and Experiments

7.1 Genre Classification Convolutional Neural Network

From the apparent link between album artwork and genre, a Convolutional Neural Network (CNN) is implemented to predict the genre of a track from the image associated with it. A CNN uses filters which convolve over an input image to create a feature map. Activation functions like ReLu are exploited alongside pooling techniques to further filter the feature map. The overall aim is to reduce the number of input nodes by capturing important information from an image [14].

For this classification task, a pre-trained model: MobileNet V2 is deployed. Depth-wise convolutions are used with this model, therefore, filters are applied to each input channel separately. Furthermore, this model has already been trained on 1.4M images with 1000 classes [36].

For this classification task, the train dataset contains 11,666 images, the test set contains 2,565 images and the validation set contains 2,575 images. The first images from the training set are shown in Figure 9.

The data is augmented by the addition of random flips to introduce diversity and reduce overfitting. Next, a base model is created from MobileNet V2 and the convolutional base layer is



Figure 9: First 15 images from the training data

frozen to ensure the weights in any given layer are not updated during training. Dimensionality reduction is performed upon the input feature map with global average pooling. The model is then compiled and trained with a base learning rate of 0.0001.

7.1.1 Evaluation Of Genre Classification CNN

After 10 epochs, the validation accuracy - the proportion of images correctly classified - is 0.28. The final validation Cross Entropy - defined as the summation of the negative log of the predicted probabilities of classification multiplied by the ground truth [14] - is 2.26. This loss function, essentially, indicates the difference in prediction to ground truth.

Figure 10 shows how the accuracy improves after each epoch. Fine tuning is used next to increase performance. This adapts the model to this specific task as the top layers of MobileNet V2 are also updated during training. The code to do so is highlighted here:

```
# also tested: fine_tune_at = 100
fine_tune_at = 120

# freeze layers prior to layer 120
for layer in base_model.layers[:fine_tune_at]:
    layer.trainable = False

# recompile model with a lower learning rate
model.compile(optimizer=tf.keras.optimizers.RMSprop(learning_rate=base_learning_rate/10),
              loss=tf.keras.losses.SparseCategoricalCrossentropy(),
              metrics=[tf.keras.metrics.SparseCategoricalAccuracy(name='accuracy')])

model.summary()

fine_tune_epochs = 10
total_epochs = initial_epochs + fine_tune_epochs
```

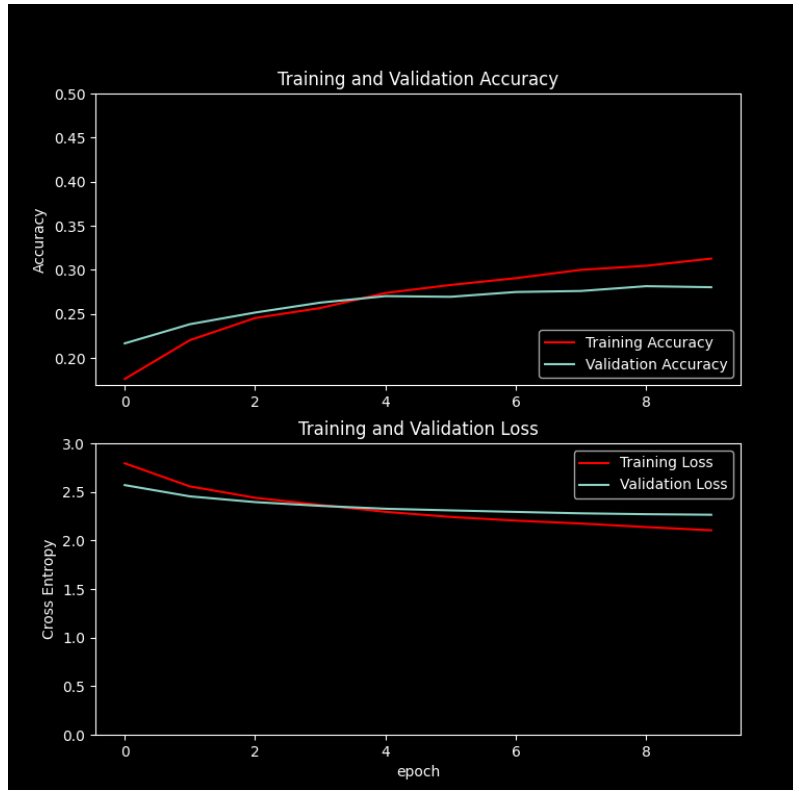


Figure 10: Training and Validation Accuracy and Loss

```
history_fine = model.fit(train_dataset,
                        epochs=total_epochs,
                        initial_epoch=history.epoch[-1],
                        validation_data=val_dataset)
```

Different experiments are carried out to discern any changes or improvements to test accuracy for this model. Specifically, the layers from which to fine tune the model as well as the pooling techniques are changed. Table 8 shows the results. It is clear to see that Global Average Pooling, alongside fine tuning at layer 120, produced the best test accuracy.

Fine Tuning At	Pooling Method	Test Accuracy
120	Average Pooling	0.305
100	Average Pooling	0.299
120	Max Pooling	0.273

Table 8: Fine Tuning and Pooling Experiments

A confusion matrix for this model is displayed in Figure 11. Finally the F1 score is computed to be 0.277 which showcases the precision and recall of the model.

The test accuracy, F1 score and confusion matrix patently illustrate how poorly this model performs. This indicates that album artwork is not a viable feature to predict genre for this specific dataset. However, based on the findings in Section 6.4, there is potential for colour to be an indicator. Moreover, there is definitely a consumer need for the incorporation of colour within MRS which is exemplified in Section 4.8, with many playlists on Spotify outlining the colours of album artwork.

7.2 Colour-Based Genre Recommender

A Colour-Based Recommender is created to fulfil this consumer requirement with genre and colour of album artwork as inputs. Dominant colours from album artwork are extracted using `colorthief` [13] and the euclidean distance between the RGB representation of the colour and an input colour

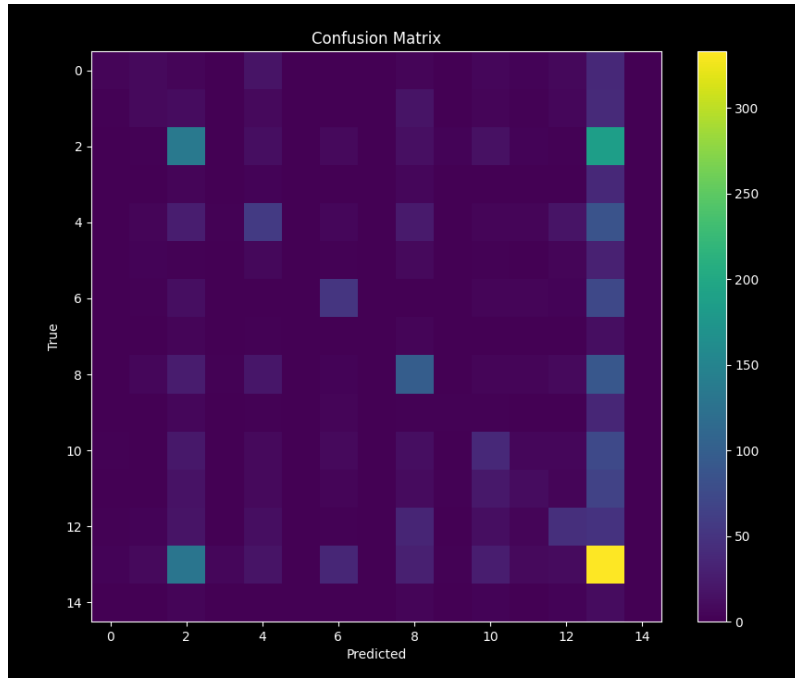


Figure 11: Confusion Matrix for Image Genre Classification

is calculated. The distances are sorted in ascending order to ascertain the most similar colours. The code to do so is shown here:

```
def recommend_tracks_colour(input_colour, genre, df):
    """
    Function to recommend songs based on the colour of the album artwork
    """
    # filter the dataframe by genre
    image_genre_df = df[df['Label'] == genre]
    # calculate the euclidean distances between the input color
    # and the colors in the dataset
    input_colour_array = np.array(input_colour).reshape(1, -1)
    genre_colours = image_genre_df[['R', 'G', 'B']].values
    distances = euclidean_distances(input_colour_array, genre_colours).flatten()
    # add distances to the genre dataframe
    image_genre_df = image_genre_df.copy()
    image_genre_df['distance'] = distances
    # sort by distance
    close_songs = image_genre_df.sort_values('distance').head(10)

    return close_songs[['track_id', 'R', 'G', 'B', 'distance']]
```

The results of this are displayed in Figure 12. The example here was the light blue input colour, alongside Jazz songs. This clearly results in an “cool” playlist automation that is missing from applications like Spotify. Even though there is an obvious consumer desire for this function [37].

7.3 Content-Based Recommender Systems

7.3.1 Tag-Based Content Recommender Using TF-IDF

The Last.fm Dataset Tags can also be used as a feature for a Content-Based Recommender System. As discussed within the Literature Review, a Content-Based Recommender System leverages the characteristics of each item and computes the similarity of the items. Finally, the MRS makes recommendations based on this [8].

The first step is to use information retrieval to extract meaning from tags. This was done with the fuzzy matching method explained in Section 6.2. To ensure the algorithms performed

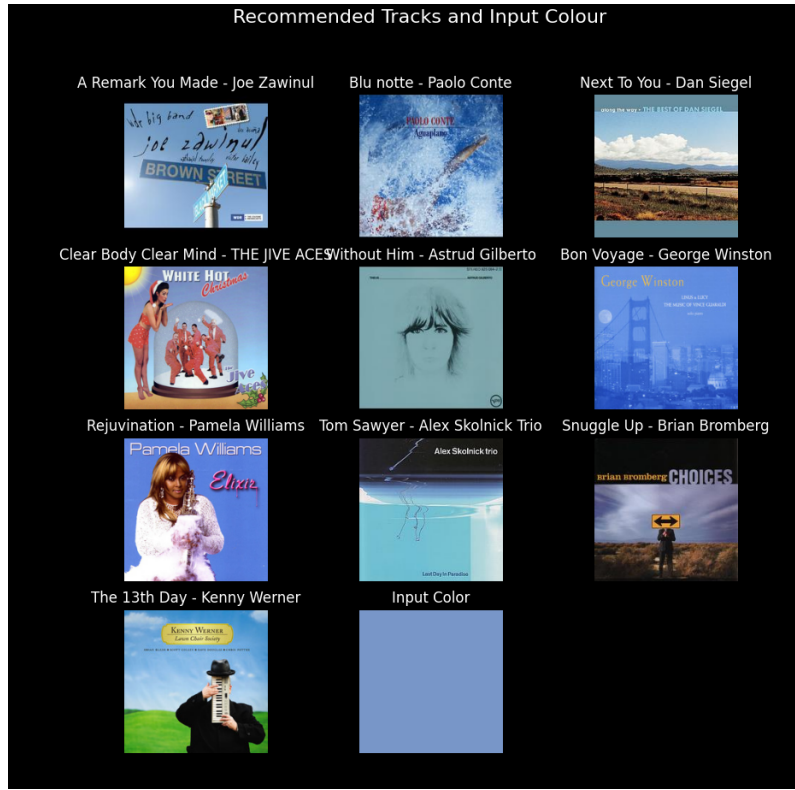


Figure 12: Recommended Songs with Similar Colour Album Artwork

are not too computationally taxing, the data is pruned. The tags are combined for each track and preprocessed, stopwords like “the”, “is” and “and” are removed. Furthermore, duplicates are found in the `artist_name` and `title` column so these are removed. NaNs are dropped for either `country`, `gender`, `language`, `religion` or `continent`. Moreover, the artist’s names and track titles are converted to lowercase so that the duplicates dropped are case insensitive. Decade is also calculated from the `year` attribute.

Then the `scikit-learn` [27] package is used with the `TfidfVectorizer` class to produce a Term Frequency-Inverse Document Frequency (TF-IDF) matrix.

This matrix contains columns representing the words in all tags and which song tags they appear in. It encapsulates the frequency of words and weights them based on how many songs they appear in. Therefore, if most songs have the tag “vocalist” then it considers this tag less important.

Then the cosine similarity is computed. This is described as:

$$\text{Cosine}(x, y) = \frac{x \cdot y}{|x||y|}$$

This metric takes two vectors that depict the TF-IDF transformation of the tags. The value ranges from -1 to 1 where 1 is an identical text and -1 is a completely different one [35]. The index of a track is retrieved, as well as a list of tuples for the cosine similarities for a particular track with all other tracks. The list is sorted from most to least similar tags and the top 25 elements are used as recommendations. So for example, for the track “In Our Dreams” by Paula Cole. The recommendations are in Figure 13:

7.3.2 Tag-Based Content Recommender Using K-Means Clustering

For the next Content-Based Recommender experiment, the artist’s name, and genre are joined and added to the tags to create a “soup” of key words. Then a `CountVectorizer()` [27] is used instead of TF-IDF. As discussed in 7.3.1 the Term Frequency - Inverse Document Frequency down-weights words that are used many times. However, when adding genre, Pop music does not become a less important feature due to the phrase being mentioned more frequently [2]. Therefore, it is less appropriate to use TF-IDF and so a `CountVectorizer()` is implemented instead.

Next, to increase the diversity of the attributes: [‘country’, ‘gender’, ‘language’, ‘religion’, ‘continent’, ‘decade’], a K-Means clustering model is carried out. The idea

track_id	title	artist_name	country	gender	language	religion	continent	genre	total_play_count	decade
TRWVVI128E07819F5	monochrome	the sundays	NaN	female	NaN	NaN	NaN	NaN	161.0	1990
TRNBFRU128F145FA91	pretender got my heart	alisha's attic	NaN	female	NaN	NaN	NaN	NaN	NaN	2000
TRLHBAU128F145E395	strangers eyes	vanessa williams	NaN	female	NaN	NaN	NaN	RnB	36.0	1990
TRGFUBM128F145C5CD	come as you are	beverley knight	NaN	female	NaN	NaN	NaN	RnB	517.0	2000
TRWQDME128F4291D10	je suis de celles	benabar	france	NaN	french	NaN	NaN	NaN	NaN	2000
TRHDLCF128F931D96E	float	sissy wish	NaN	female	NaN	NaN	NaN	Pop_Rock	5.0	2000
TRLXSD128F423CF70	na nighty	remi nicole	NaN	female	NaN	NaN	NaN	NaN	8.0	2000
TRGAOJF128F424112D	wooden heart	the duke spirit	NaN	female	NaN	NaN	NaN	Pop_Rock	244.0	2000
TRXQWY012903D12C27	i've got a secret	kate nash	NaN	female	NaN	NaN	NaN	Pop_Rock	1177.0	2010
TRGQURV128E078286E	into dust	mazzy star	NaN	female	NaN	NaN	NaN	Pop_Rock	2637.0	1990
TRIBEHM128F42746DA	miej oczy otwarte	reni jusis	NaN	female	NaN	NaN	NaN	Electronic	4.0	1990
TRLPMWF128F427858C	les vidanges du diable	francis cabrel	france	NaN	french	NaN	NaN	Pop_Rock	NaN	1990
TRTVBYR128F421F6D4	om shiva	zion train	world	NaN	NaN	NaN	NaN	Electronic	8.0	1990
TRTKQVR12903CD6CEE	all i want	sarah blasko	NaN	female	NaN	NaN	NaN	NaN	722.0	2000
TRIMZNN128F4288E59	the monochrome set	the monochrome set	united kingdom	NaN	NaN	NaN	NaN	Pop_Rock	85.0	1970
TRCERBN128F423C605	memories (album version)	eisley	NaN	female	NaN	NaN	NaN	Pop_Rock	NaN	2000
TRIWXF0128F92CA40E	don't give up on us	sarah mclachlan	NaN	female	NaN	NaN	NaN	Pop_Rock	922.0	2000
TRUMQSA128F931D475	time	chantal kreviazuk	NaN	female	NaN	NaN	NaN	NaN	346.0	2000
TRAQUB0128F429E986	he turns down	cat power	NaN	female	NaN	NaN	NaN	Pop_Rock	292.0	1990
TRARGFX128F9322288	wires	coconut records	NaN	male	NaN	NaN	NaN	NaN	344.0	2000
TRDHEVG128F423EEEE	calling me	jill cunniff	NaN	female	NaN	NaN	NaN	Pop_Rock	907.0	2000
TRPVUAL128F423A728	hometown glory (album)	adele	NaN	female	NaN	NaN	NaN	Pop_Rock	NaN	2000
TRLJCJF128F92EF1D0	hometown glory	adele	NaN	female	NaN	NaN	NaN	NaN	NaN	2000
TRARXEX12903CB1408	furthest sense	ride	united kingdom	NaN	NaN	NaN	NaN	Pop_Rock	14.0	1990

Figure 13: TF-IDF Recommendations for the song “In Our Dreams” by Paula Cole

behind this is that once several clusters are created, the recommender can choose items from different groups, thereby increasing diversity.

This algorithm initialises K random centres and assigns the nearest points to these centres as a cluster. The cluster centre points are changed based on the average of the assigned points and the algorithm is repeated until the centroids no longer change [14].

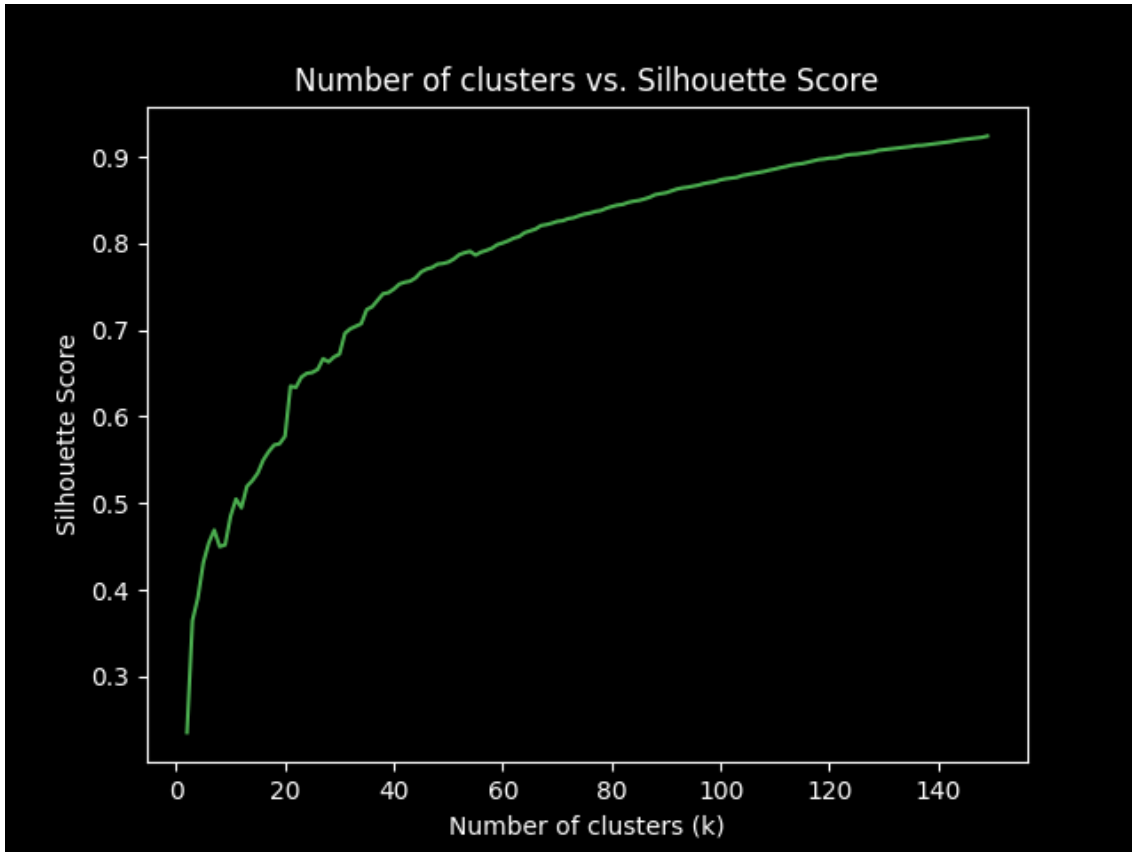


Figure 14: Silhouette Score of K-Means Model

The diversity attributes are modified using one hot encoding so that the categorical data transforms into numerical data. After this, a K-Means model is instated with different values for K ranging from 2 to 150. A silhouette score is computed for each model which is demonstrated in Figure 14. The silhouette score is

$$(b - a) / \max(a, b)$$

where a is the mean intra-cluster distance and b is the mean nearest cluster distance [14]. Here, by increasing the number of clusters the silhouette score would inherently increase, however to

prevent overfitting K is chosen as 60. A recommender function is then created using the “soup” content and recommendations are taken from unique clusters. So for the same track: “In Our Dreams” by Paula Cole, the K-Means recommendations are in Figure 15. This clearly shows the increase in diversity with many more countries being listed.

track_id	title	artist_name	country	gender	language	religion	continent	genre	total_play_count	decade
TRWMDFF128F93243EF	deixa-me rir	jorge palma	NaN	NaN	portuguese	NaN	NaN	NaN	NaN	1990
TRWMDR8128F429726A	small song	lhasa de sela	NaN	female	NaN	NaN	NaN	NaN	829.0	2000
TRFLU4U128F92F2135	dead wrong	the fray	taiwan	NaN	NaN	NaN	NaN	Pop_Rock	20.0	2000
TRWJXS128F92D2CA0	tobacco road	the nashville teens	united states	NaN	NaN	NaN	north america	NaN	NaN	1960
TRZVND128F428898F	just me and you	phyllis hyman	NaN	female	NaN	NaN	NaN	RnB	NaN	1990
TRIVSV128F4285606	me falta el silencio	estopa	spain	NaN	NaN	NaN	NaN	Latin	3007.0	1990
TRJRAV128F14589C5	life without buildings	japan	united kingdom	NaN	NaN	NaN	NaN	Pop_Rock	17.0	1980
TROUHH128F93093CD	demoliendo hoteles	charly garcia	NaN	NaN	NaN	NaN	south america	NaN	NaN	1980
TROSRI128F93CB04E3	last ever lone gunman	snow patrol	united kingdom	NaN	NaN	NaN	NaN	NaN	387.0	2000
TRIQHA128F4255A67	shine	matt redman	NaN	NaN	NaN	christianity	NaN	NaN	7.0	2000
TRHZPD128F146C98E	ghanwa bossanova	natacha atlas	world	NaN	NaN	NaN	NaN	Electronic	14.0	2000
TRMLMF128F9244D9	hör weg	beginner	germany	NaN	german	NaN	NaN	Rap	19.0	2000
TRFNAI128F4226F53	organic	agoria	france	NaN	french	NaN	NaN	Electronic	17.0	2000
TRHLSIP128F932D168	vinushka	dir en grey	japan	NaN	japanese	NaN	NaN	Pop_Rock	261.0	2000
TRJZLR128F1458507	jailhouse rock	the jeff beck group	united kingdom	NaN	NaN	NaN	NaN	NaN	21.0	1960
TRLDHB128F9310E94	i'm going down	the rolling stones	united kingdom	NaN	NaN	NaN	NaN	Pop_Rock	94.0	1970
TRVHIUI128F147A337	girl (you chew my mind up)	the presets	NaN	NaN	NaN	NaN	australia	Pop_Rock	250.0	2000
TRCOUN128F42634A6	bucci bag	andrea doria	italy	NaN	italian	NaN	NaN	Electronic	NaN	2000
TRZROXB128F92D0438	palmitos park	el guincho	spain	NaN	spanish	NaN	NaN	Pop_Rock	795.0	2000
TRWJUMY128F93CB308	you're not sorry	taylor swift	united states	NaN	NaN	NaN	NaN	NaN	NaN	2000
TRBAMPN128F9339706	body and soul	mai tai	NaN	female	NaN	NaN	NaN	Pop_Rock	NaN	1980
TRIPIRY128F4290F32	money love	neneh cherry	united kingdom	NaN	NaN	NaN	NaN	RnB	71.0	1990
TRZHAEP128F426958D	be me	ayria	canada	NaN	NaN	NaN	NaN	Electronic	97.0	2000
TRPKCZS128F92E3085	motel	telefunka	NaN	male	NaN	NaN	NaN	NaN	110.0	2000
TRJHZFL128F4276653	adoro	armando manzanero	NaN	NaN	NaN	NaN	south america	Latin	NaN	2000

Figure 15: K-Means Recommendations for the song “In Our Dreams” by Paula Cole

7.3.3 Evaluation of Tag-Based Content Recommenders

To evaluate both of these Content-Based recommender systems, an `average_diversity_score` is calculated for the attributes indicating diversity: gender, language, religion, continent, country and decade. For instance, for the track “forest for the trees by kemopetrol”, the 25 song recommendations have 1 gender label, 4 different language labels, 1 labelled continent, 9 countries and 2 decades. Therefore, the diversity score for language is 4.

10 tracks are randomly selected to provide 25 recommendations and the average score for each diversity attribute is calculated. This is done for the TF-IDF and Clustering recommender. Naturally, this method would be improved if all tracks had labels - but since the information was retrieved from the tags there are not as many and some are NaNs. The code snippet for the average diversity score is shown below:

```
average_diversity_score = {attribute: 0 for attribute in attributes}
for _, diversity_score in diversity_results:
    for attribute in attributes:
        average_diversity_score[attribute] += diversity_score[attribute]

average_diversity_score = {attribute: score / num_samples for attribute,
                           score in average_diversity_score.items()}
```

The outcome of the average diversity score for each attribute is displayed in Table 9. From this, it is clear that the Clustering approach results in greater diversity from the chosen attributes. Specifically, the Country feature which displays an average of 10.8 different countries for the 10 sets of 25 recommendations given.

Category	Average Diversity Score (K-Means Clustering)	Average Diversity Score (TF-IDF)
Gender	2.0	1.3
Language	5.8	1.0
Religion	1.1	0.0
Continent	3.1	0.7
Country	10.8	3.6
Decade	5.4	3.2

Table 9: Average Diversity Scores

7.4 Collaborative Filtering Recommender

As discussed in Section 4.1, in User Based Collaborative Filtering, similar users are grouped together and the system predicts a new rating from a user that has not yet listened to or rated a track. This differs from Item Based Collaborative Filters where similar items are grouped together and a new item is recommended to a user based on their previous ratings of their tracks. User based filtering is used here as it is more likely to give interesting and unexpected recommendations.

To start with, the Train Triplets from Section 5.1.3 - which includes users and their listening history - is pruned. The mean number of times that a user listens to any one song is 2.65, the maximum number is 1467 and the minimum is 1. Furthermore, 486,546 tracks have been listened to by a single user only once.

The first test method to filter the data is to use only values within the interquartile range (IQR). The interquartile range for play count per user is from 1-3 songs. Excluding songs that have been listened to more than 3 times would be skewing the data greatly. In the section 7.4.1 different values for minimum listens are experimented with.

Moreover, users that are mostly inactive are removed. This encapsulates users who have only listened to a certain number of songs overall, again this is experimented with in 7.4.1.

Then a ranking of 1 to 5 is applied to each song based on how many times they have listened to it. The normalisation method is min max feature scaling [18]. This changes the implicit feedback of play count into explicit feedback [2]. However, predicting ranking and predicting play count are also compared in the section 7.4.1.

The data is formatted and pivoted into a matrix so that unique tracks make up the columns of the DataFrame and users make up the rows. The sparsity of this matrix is defined as the number of empty cells as a proportion of the total number of cells. The sparsity for this matrix is 99.89%. Due to this sparsity, a simple K-Nearest Neighbours Model would not work as it relies heavily on distance metrics which would be dominated by the zero entries. Therefore, matrix factorisation can be utilised. This fascinating methodology decomposes the 33,943 by 2611 matrix into two factors, one with 33,943 rows which represent the users and one with 2611 columns which represent the tracks. To obtain these factors, Singular Value Decomposition is executed.

Singular Value Decomposition (SVD) is a matrix factorisation approach that takes the form: $A_{mn} = U_{mm}\Sigma_{mn}V_{nn}^T$. Where U is orthogonal such that $U^T U$ is the identity matrix with ones on the diagonal and zeros elsewhere. The columns of U are orthonormal eigenvectors of AA^T , the columns of V are orthonormal eigenvectors of $A^T A$ and Σ is a diagonal matrix which holds the square root of the eigenvalues from U or V [4].

To ensure the data is not biased in any way, the NaNs are filled with 0s once the data is centred using the average ratings from a user. The SVD is performed using the library SciPy [38].

```
U, sigma, Vt = svds(user_ratings_sparse)
```

Where `user_ratings_sparse` is the sparse matrix of user ratings. After the decomposition, the dot product between U , Σ and V^T is computed to reconstruct the ratings matrix and predict ratings for tracks that users have not yet listened to.

7.4.1 Evaluation of Collaborative Filtering Recommender

The data is split into test and train sets in preparation for evaluation. The test set is 20% of the data and the Root Mean Squared Error (RMSE) is calculated. The RMSE measures the difference between the predicted values and the actual values for ratings [14]. The mean absolute error (MAE) is also calculated for this model. Moreover, a baseline model is created which populates the predictions with the user's average rating. This baseline model is used for comparison. As well as this, models predicting play count and rating are experimented with. Furthermore, changes to the threshold for minimum play count and minimum played tracks per user are tested.

7.4.2 Results of Collaborative Filtering Recommender

As demonstrated in Table 10, the Play Count Models performed poorly in comparison to the Rating Prediction Models. Furthermore, when the minimum play count was increased the models performed generally worse. The lowest error came from the Rating model with minimum play count as 3 and 2 tracks per user. This implies that the predicted values for this model were the closest to the actual values. This model is chosen as the final model. The worst performing model was the Play Count Prediction Model where min play count was 5 and min played tracks per user was 15. This actually was outperformed by the baseline model which assigns the average.

Features						
Min Play Count	3	5	5	10	5	3
Min Played Tracks Per User	2	5	10	5	15	20
Rating Prediction Model						
MAE	0.0182	0.0213	0.1504	0.0613	0.6152	0.5156
RMSE	0.0385	0.0429	0.2411	0.0613	0.8690	0.7789
Play Count Prediction Model						
MAE	4.0401	4.1032	3.3465	5.5856	14.8663	11.2886
RMSE	9.8844	8.2694	5.3642	11.7301	18.4076	15.3592
Rating Prediction Model - Baseline						
MAE	0.0209	0.0213	0.1530	0.0290	0.2300	0.1637
RMSE	0.0512	0.0430	0.2432	0.0609	0.4365	0.3269
Play Count Prediction Model - Baseline						
MAE	4.0393	4.1102	3.4037	5.5554	3.8529	2.8231
RMSE	9.8894	8.2877	5.4110	11.6709	7.3111	5.6384

Table 10: Comparison of Metrics Across Different Configurations

Finally, to add the diversification element, a function is created which, when given a user, employs the Collaborative Filtering Model to output 15 songs that contain the highest predicted ratings. These songs are then sorted by lowest play count and the recommender takes the top 5 of these. This ranking based on low play count ensures the recommendations are both relevant and obscure. This code is shown below:

```
def recommend_tracks(user, train_df, calc_pred_ratings_df, collab_df):
    '''
    function to recommend tracks with lower play count but still with the
    highest predicted ratings
    '''
    # tracks that the user has already rated
    user Rated_tracks = train_df.loc[user,:].dropna().index
    # predicted ratings for tracks the user has not rated
    predicted_ratings = calc_pred_ratings_df.loc[user].drop(user Rated_tracks,
                                                             errors='ignore')

    # sort ratings
    sorted_predicted_ratings =
        predicted_ratings.sort_values(ascending=False).head(15)
    # join predictions with play count
    recommendations_df = pd.merge(sorted_predicted_ratings,
                                   collab_df[['track_id',
                                                'total_play_count']], on='track_id',
                                   how='left').drop_duplicates()

    # take the top 5 of the lowest played songs
    sorted_recommendations =
        recommendations_df.sort_values(by=['total_play_count'],
                                         ascending=True).head(5)

    return sorted_recommendations
```

For the user: "eb94d439dd9b75af60cc704e116d06ac1dcbefa8" with the tracks ratings in Figure 16. The are recommended the songs found in Figure 17

8 Conclusions & Future work

In conclusion, this research has successfully produced three recommender systems with the main focus area being increasing diversity by considering cultural, obscure and visual attributes to do so. The objectives in the Section 3.3 outline how this research will investigate content-based and

	track_id	eb94d439dd9b75af60cc704e116d06ac1dcbefa8	title	artist_name
0	TRMCERZ128F425EA28	1.010349	Take A Bow	Leona Lewis
1	TRPJRSO128F425EA31	1.010349	I'm You	Leona Lewis
2	TRUBEDK128F42B01C6	1.020699	I Didn't Mean To	Heather Headley

Figure 16: Tracks already rated by User: “eb94d439dd9b75af60cc704e116d06ac1dcbefa8”

	track_id	eb94d439dd9b75af60cc704e116d06ac1dcbefa8	total_play_count	title	artist_name
0	TRWMDY128E0784323	1.013801	762.0	Genocide	Thin Lizzy
1	TRPESQP128F92F12C7	1.013799	769.0	Quiet	Rachael Yamagata
2	TRDHEVG128F423EEEE	1.013799	907.0	Calling Me	Jill Cunliff
3	TRJMLCS128F146FE4C	1.013799	1053.0	Na Estrada	Marisa Monte
4	TRWYNJA128F1467C1A	1.013799	1189.0	Talk (Junkie XL Remix)	Coldplay

Figure 17: Recommended Songs for User: “eb94d439dd9b75af60cc704e116d06ac1dcbefa8”

collaborative filtering recommender systems, with the priority being exploring the untapped long-tail. The Tag-Based Content Recommender with K-Means clustering successfully improved the `average_diversity_score` with recommendations bolstering different languages, countries and continents. The Collaborative Filtering Recommender also showed success in predicting user preferences by leveraging matrix factorisation techniques.

This work is not yet finished as it is clear that the CNN image classifier was limited in the final accuracy. The next steps here would be to create a genre predictor based on dominant colour as there were some significant findings in how the colours used in album artwork could be linked with genre. Also, the aim of creating an app or website was not met due to constraints in timing. Within this research it was not deemed as the main priority.

The Collaborative Filtering Recommender could be further refined by evaluating how obscure the song recommendations are in comparison to an MRS that does not rank tracks based on play count. Moreover, a K-Nearest Neighbour Model could be experimented with to confirm the most appropriate model.

Additionally, a hybrid approach could combine these separate recommender systems into a cohesive MRS.

Upon reflection, it would be useful to partner with an actual Music Streaming Platform to obtain music with features readily available. Using information retrieval to pull out country, gender, religion and continents introduced inaccuracies as there was no ground truth for the entire dataset. Additionally, the MSD contains tracks released up until 2011 and so more up to date music would ensure the MRS reflects the current music scene. However, this research provides a solid foundation for different techniques to diversify MRS and explore the long-tail. This research has also analysed the many aspects of computing taste and the variables that go into this, including artistic and cultural.

Future work in this area should aim to utilise real time data with more attributes to explore. One aspect of this could be how being an immigrant could influence music taste and the desire to connect to cultural roots through music. As well as this, the mood of an individual and the impact this has on what users listen to could also be investigated.

In terms of the gender exploration, more robust data could give insights into the biases of MRS and whether certain genres can be de-genderised through the techniques employed in this research. Also, further work should include LGBTQIA+ attributes and non-binary artists.

Regarding social tagging, more work in this area could investigate why individuals tag the things they do: why a user may be more likely to tag a female artist than a male, or Spanish over English. This would be enlightening in terms of showing what consumers really care about within their music and what they value.

Finally, it would be gainful to create an application wherein users can choose the attribute they want to diversify. The MRS created in this research clustered the attributes together which takes the power out of the user’s hand. It would be interesting to have an option to select “diversify decades” and have an MRS produce similar tracks but from different years.

In summary, there are many avenues for exploration and this research provides a solid foundation for this.

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